

GROUIN DE CANCALE, France

WMO: 071270

Lat: 48.7098N

Lon: 1.8451W

Elev: 135

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.4	31.1	19.7	14.9	30.0	24.4	18.6	33.9	51.8	47.6	46.6	46.6	14.2	100	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	74.9	66.9	72.3	65.6	70.2	64.6	68.3	72.7	66.8	70.8	65.5	69.1	10.9	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	98.4	70.6	65.3	94.1	69.1	64.0	89.7	67.7	32.6	73.0	31.4	70.7	30.4	69.5	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
44.3	37.1	31.5	DB	23.5	82.0	5.4	4.0	19.6	84.9	16.5	87.2	13.4	89.5	9.5	92.4	
			WB	21.7	70.8	5.1	1.6	18.0	72.0	15.0	72.9	12.2	73.8	8.5	75.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	43.4	43.8	47.4	49.2	55.5	59.6	63.7	64.9	60.6	55.7	49.1	45.2
	DBStd	8.82	6.52	5.97	4.16	4.07	4.34	3.95	3.36	3.53	3.71	4.21	5.39	5.73
	HDD50	805	212	182	99	60	5	0	0	0	0	6	75	166
	HDD65	4380	671	595	545	473	295	171	67	44	139	287	478	615
	CDD50	1987	7	7	19	37	176	288	424	463	319	184	47	16
	CDD65	86	0	0	0	0	1	9	26	42	8	0	0	0
	CDH74	131	0	0	0	0	3	29	31	62	6	0	0	0
	CDH80	12	0	0	0	0	0	4	2	6	0	0	0	0
Wind	WSAvg	15.2	16.5	15.7	15.3	15.9	13.2	14.3	13.5	13.3	15.2	16.4	16.2	17.0
Precipitation	PrecAvg	28.0	2.4	2.1	1.7	2.1	2.0	1.8	1.8	2.0	2.4	3.3	3.1	3.2
	PrecMax	35.6	5.1	4.8	5.4	4.4	4.7	3.9	3.7	4.6	5.4	5.4	7.0	7.0
	PrecMin	21.7	0.4	0.4	0.4	0.7	0.6	0.6	0.7	0.6	0.6	0.8	1.4	0.9
	PrecStd	4.0	1.5	1.1	1.1	0.9	1.0	1.0	0.8	1.1	1.3	1.1	1.4	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	57.9	60.6	66.2	71.8	77.8	77.9	79.8	73.4	68.9	60.8	58.4
		MCWB	53.6	54.4	55.0	57.4	62.3	67.0	66.6	68.9	66.3	62.6	58.6	56.3
	2%	DB	54.8	55.1	57.4	60.7	67.2	71.9	73.8	75.2	70.3	65.2	58.5	56.5
		MCWB	52.7	51.9	53.7	54.9	59.7	63.8	66.6	67.7	64.7	61.2	56.4	54.0
	5%	DB	53.1	53.2	54.9	57.7	64.6	67.8	71.6	72.3	68.2	63.2	56.9	54.8
		MCWB	51.1	50.8	51.8	53.4	58.6	62.3	65.7	66.1	63.2	59.9	55.0	52.6
	10%	DB	51.0	51.3	53.0	55.1	62.1	65.2	69.3	70.2	66.3	61.5	55.7	52.9
		MCWB	48.9	49.2	50.3	51.8	57.4	60.7	64.2	65.0	62.0	58.8	53.8	50.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	54.8	56.1	58.7	63.7	68.0	69.6	70.4	68.7	63.6	59.0	56.9
		MCDB	55.7	57.1	59.4	63.3	70.0	75.5	73.4	76.7	71.9	66.3	60.0	58.0
	2%	WB	53.0	52.8	54.0	55.8	60.9	64.9	67.5	68.7	65.9	61.9	56.8	54.6
		MCDB	54.5	54.4	56.5	59.7	65.3	70.6	71.9	73.4	68.3	64.3	57.9	56.0
	5%	WB	51.3	51.1	52.4	53.9	59.2	62.9	66.1	67.3	64.4	60.4	55.5	52.9
		MCDB	52.9	52.8	54.5	56.8	63.6	67.1	70.3	71.0	67.0	62.5	56.7	54.6
	10%	WB	49.2	49.5	50.8	52.1	57.7	61.2	64.8	66.0	62.8	59.3	54.2	51.0
		MCDB	50.7	51.2	52.7	54.4	61.7	64.7	68.7	69.4	65.9	61.2	55.5	52.6
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.9	7.3	7.6	8.3	8.1	7.9	8.1	7.6	6.8	6.2	6.3
		MCDBR	7.7	9.4	10.3	11.4	12.3	14.0	11.6	11.4	10.4	8.9	6.7	7.5
		MCWBR	7.1	7.8	7.3	7.5	7.4	7.7	7.1	6.2	6.4	6.4	6.1	7.0
	5% WB	MCDBR	7.5	9.1	9.5	9.8	11.0	12.7	10.4	10.1	9.1	8.0	6.6	7.5
		MCWBR	7.4	8.0	7.5	7.2	7.1	7.6	7.1	6.1	6.7	6.5	6.3	7.3
Clear-Sky Solar Irradiance	taub	0.330	0.346	0.376	0.391	0.397	0.394	0.391	0.382	0.368	0.359	0.335	0.323	
	taud	2.424	2.395	2.314	2.290	2.317	2.359	2.368	2.414	2.434	2.451	2.443	2.396	
	Ebn at Noon	231	254	264	272	274	275	274	272	265	246	226	215	
	Edh at Noon	21	27	34	38	39	38	37	34	30	25	21	19	
All-Sky Solar Radiation	RadAvg	304	550	901	1384	1638	1811	1734	1479	1161	666	371	254	
	RadStd	28	64	101	143	138	193	114	120	87	53	39	32	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page